Tide retrospective: the pairwise jet difference limit (weighted-jet programme, stages 3A–3C)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

The weighted-jet programme's original plan built the full weighted exponential expansion: multi-index coefficient polynomials, a triangular decomposition, and a normalized quotient recurrence. This tide replaced that middle of the programme wholesale. The observation: recovery is a *comparison* statement, so each rung needs only the first-order difference of two jet Gibbs weights — never the full expansion of either. A dedicated consult endorsed the replacement ("mathematically sound, and the shortest route") and contributed two refinements that shaped every proof in the file: the exponential *secant* bound in place of a Taylor bound with an $e^{|D|}$ factor that the envelopes cannot control, and the observation that $\rho = \min(\rho_1, \rho_2)$ envelopes both jets and the entire segment between them, so no auxiliary "mixed jet" is needed.

2 The thing formalised

Nine declarations in `Laplace/OneD/JetDifference.lean`, sorry-free with zero warnings. For two enveloped jets agreeing below rung $r = i_0 + 1$, the potential difference factors as $L_q^1 - L_q^2 = q^r g(q)$ with g polynomial in q and $g(0) = (c^1_{i_0} - c^2_{i_0})\, u^{2k+r}$; the pointwise limit

$$\frac{e^{-L_q^1(u)} - e^{-L_q^2(u)}}{q^r} \xrightarrow[q \to 0^+]{} -(c^1_{i_0} - c^2_{i_0})\, u^{2k+r}\, e^{-au^{2k}}$$

upgrades, by dominated convergence along $\mathcal{N}[>]\,0$, to the programme's reusable core[1]:

$$\frac{J_s^1(q) - J_s^2(q)}{q^r} \longrightarrow -(c^1_{i_0} - c^2_{i_0}) \int_{\mathbb{R}} u^{s+2k+r}\, e^{-au^{2k}}\, du.$$

Supporting cast: the secant bound $|e^{-x} - e^{-y}| \le |x - y| \max(e^{-x}, e^{-y})$, the min-envelope lemma, the absolute-value monomial-Gibbs integrability, and the q-free majorant $\sum_j |\delta_j|\,|u|^{s+2k+j}\, e^{-\rho u^{2k}}$ valid on $0 < q \le 1$.

3 Proof strategy

The pointwise limit avoids the removable singularity of $(e^{-D} - 1)/D$ entirely: writing $e^{-D} - 1 = -D + E_2(D)$ with the seabed's second-order exponential remainder, the remainder term is squeezed

[1] `jet_difference_integral_limit`.

against $q^r|g(q)|^2 \max(1, e^{-D})/2$, which is continuous with value 0 at $q = 0$; everything else is a product of continuous functions evaluated along $\mathcal{N}[>]\,0$. The dominated-convergence stage — flagged by the consult as the likely bottleneck — went through on the first full build, using Mathlib's filter version of the dominated convergence theorem, with `gcongr` carrying the division-monotone calc steps.

The numerical check (executed before the consult returned) verified the normalized form of the limit at $k = 1$: measured over predicted ratios 0.9999 at $q = 0.005$ for $s \in \{1, 3\}$, and the $s = 2$ coefficient vanishing by parity — confirming the consult's warning that the recovery data must include the matching moment $s = 2k + r$.

4 Roadblocks and resolutions

Genuinely smooth for a 415-line file: two build iterations. The first draft used the q-continuity lemma where u-continuity was needed (caught immediately), and the majorant algebra wanted one extra calc step separating the q^r cancellation from the sum reshaping. One reusable observation: the natural-subtraction exponents $j - i_0$ in the factorization are harmless precisely because the $j < i_0$ terms carry $\delta_j = 0$, so both sides vanish there regardless of what the truncated exponent does.

5 The deliberation

The route consult is archived verbatim in the tide log.[2] Its decisive content was again mathematical: the secant-versus-Taylor correction (the drafted $e^{|D|}$ bound "is not, by itself, controlled by separate lower envelopes"), and the segment-envelope simplification. It also restructured the remaining programme into stages 3A–3G, of which this tide closes 3A–3C; the eliminated stages (multi-index P_n combinatorics, quotient recurrences) are noted as worthwhile only if explicit expansion coefficients are ever needed for their own sake.

6 Mathlib vs new

Mathlib lookup: the filter dominated convergence theorem, the neighborhood-filter membership for $(0, 1]$, and the standard continuity/integrability combinators. Seabed reuse: the stage-2 jet machinery wholesale, and `ExpRemainderSigned`'s two-sided remainder from the gamma-rung programme. New: the secant bound, the factorization, and the two limit theorems.

7 What the next tide inherits

Stages 3D–3G: the quotient-difference lemma (limits of $A_1/B_1 - A_2/B_2$ from limits of scaled differences), the one-rung recovery ($s = 2k + r$ turns the covariance into stage 1's positive variance), the finite strong induction over rungs, and the transfer back to t-data through the stage-2 scaling identity. All four are now bookkeeping over this tide's limit theorem; they plausibly fit in a single closing tide.

[2] `tide-log/gpt56_jet_difference_route_v1.md`.